Pseudocode Questions

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What operation does the following pseudocode perform?

0	1	2	3	4	5
A	d	a	r	s	h

1. Declare a String variable "word"
   Declare a variable i
   Take a string as input in the word
   for i = 0 to 0
       print word[i]
   End for
   Algorithm End

String word = "Adarsh"; int i;

i = 0 to 0

word[0] = 'A'

A. It prints the last character of the string

B. It prints characters in increasing order

C. It prints the first character of the string

D. It reverses the string

What operation does the following pseudocode perform?

2. Calculate the output of the following pseudocode for input p = 3, q = 8, r = 1.

```
Integer p, q, r, sum
Read p, q, r
Set sum = p + q + r
if ((p NOT EQUALS 0) and (sum EQUALS 11) and (q EQUALS 4) and (r NOT
EQUALS 0))
    Print " Success"
Otherwise
    Print "Fail"
End if
```

A. Success B. Fail C. Error D. None

p = 3, q = 8, r = 1, sum = p + q + r = 3 + 8 + 1 = 12

if ((true) AND (false) AND (false) AND (true))

false

What operation does the following pseudocode perform?

```
3.  Integer arr[5] = {2,4,6,8,10}
    Integer sum = 0
    For i = 0 to 4
        If arr[i] mod 4 == 0 then
            sum += arr[i]
        End if
    End for
```

A. 12

B. 22

C. 59

D. Don't know

sum = 0 4 12

arr[5] =

2	4	6	8	10
0	1	2	3	4

sum = sum + arr[i]

for i = 0 ⇒ arr[0] % 4 ⇒ 2 % 4 = 2 ≠ 0

for i = 1 ⇒ 4 % 4 = 0 == 0 ✓

for i = 2 ⇒ 6 % 4 = 2 ≠ 0

for i = 3 ⇒ 8 % 4 = 0 == 0 ✓

for i = 4 ⇒ 10 % 4 = 2 ≠ 0

What operation does the following pseudocode perform?

4.
```
Integer x, y, z
Set x = 10, y = 16, z = 3
if(x > y)
    x = y
Else
    y = x
End if
if(z > y)
    z = y
Else
    y = z
End if
Print x + y + z
```

$x = 10$, $y = 16$, $z = 3$

$y = 10$

$y = 3$

10 > 16 false

3 > 16 false

$x + y + z = 10 + 3 + 3$

$= 16$

A. 13 B. 12 C. 20 D. 16

What operation does the following pseudocode perform?

5. Print the output of the following pseudocode for x = 9, y = 7.

```
Integer funn(Integer x, Integer y)
Integer z
Set z = 2
y = y mod z
x = x mod z
return x + y
End function funn()
```

A. 2 B. 3 C. 17 D. 5

$x = 9, \quad y = 7$

funn (x, y) $\quad z = 2$

funn (9, 7)

$y = y \% 2 = 7 \% 2 = 1$

$x = x \% 2 = 9 \% 2 = 1$

$x + y = 1 + 1 = 2$

2) 7 (3
6
1

2) 9 (4
8
1

✓ Bitwise XOR ⇒ Return result = 1 only if bits are different.

✓ Bitwise AND ⇒ Return result = 1 only if both bits are 1.

What operation does the following pseudocode perform?

6.
```
Integer x, y, z
Set x = -2, y = 3, z = 1
if(x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2))
    x = x - 2
    y = x
Else
    x = z
    y = y ^ 2
End if
Print x+ y+ z
```

$x = -2, \quad y = 3, \quad z = 1$

& = Bitwise AND
&& = Logical AND
^ = Bitwise XOR

$x = x - 2 \Rightarrow x = 1$

$y = y \wedge 2 \Rightarrow y = 1$

$x + y + z \Rightarrow 1 + 1 + 1 = 3$

A. 2 B. 11 C. 3 D. 0

$2 \& 2 \Rightarrow 10 \;\&\; 10 = 10 \Rightarrow 2$

$3 \& 3 \Rightarrow 11 \;\&\; 11 = 11 \Rightarrow 3$

$2 \wedge 2 = 10 \wedge 10 = 00 \Rightarrow 0$

$x + (2 \& 2) = -2 + 2 = 0$

$y + (3 \& 3) = 3 + 3 = 6$

$z + (2 \wedge 2) = 1 + 0 = 1$

$y = y \wedge 2 = 3 \wedge 2 \Rightarrow 11 \wedge 10 = 01 \Rightarrow 1$

if (0 && 6 && 1)

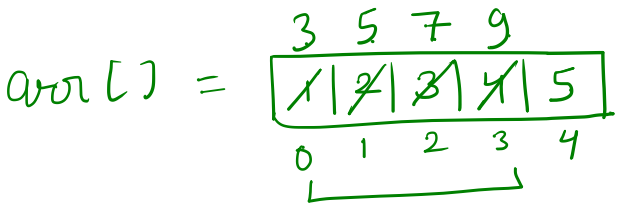
false

What operation does the following pseudocode perform?

7. What will be the output of the following pseudocode for arr[] = 1,2,3,4,5

```
Initialize i, n
Set arr[]= 1, 2, 3, 4, 5
for i = 0 to n - 2
      arr[i] = arr[i] + arr[i+1]
End for
print the array of elements
```

A. 3 3 4 9 5 B. 4 3 7 5 2
C. 3 5 7 9 5 D. 1 2 3 4 5



$i = 0 \Rightarrow arr[0] = arr[0] + arr[1] = 1 + 2 = 3$

$i = 1 \Rightarrow arr[1] = arr[1] + arr[2] = 2 + 3 = 5$

$i = 2 \Rightarrow arr[2] = 3 + 4 = 7$

$i = 3 \Rightarrow arr[3] = 4 + 5 = 9$

What operation does the following pseudocode perform?

8. Integer p, q, r, s
   Set p = 4, q = 2, r = 1
   s = (p AND q) OR (r + 1)
   Print s ⇒ 2

A. 1 B. 2
C. 3 D. 5

s = ?

p = 4, q = 2, r = 1

p AND q = 4 AND 2 ⇒

```
  1 0 0
& 0 1 0
-------
  0 0 0  ⇒ 0
```

r + 1 = 1 + 1 = 2

s = 0 OR 2 ⇒

```
  0 0
| 1 0
-------
  1 0  = 2
```

What operation does the following pseudocode perform?

9. Predict the output of the following pseudo-code

```
Integer x = 1, y = 1, z = 1
x = y * z
z = y - z
y = z / y
Print x, y, z
```

A. 1 2 1 B. 1 0 0
C. 2 1 0 D. 3 4 1

$x = 1, \quad y = 1, \quad z = 1$

$x = y * z = 1$

$z = y - z = 0$

$y = z / y = 0 / 1 = 0$

$x = 1, \quad y = 0, \quad z = 0$

What operation does the following pseudocode perform?

10. Predict the output of the following pseudo-code

```
Integer p, q, r, s
Set p = 1, q = 2, r = 0
s = (p XOR q) OR (r | 1)
Print s
```

$p = 1 , \quad q = 2 , \quad r = 0 , \quad s = ?$

$p \text{ XOR } q = p \wedge q = 1 \wedge 2 \Rightarrow$

$$\begin{array}{r} 0\ 1 \\ \wedge\ 1\ 0 \\ \hline 1\ 1 \end{array} \Rightarrow 3$$

$r \mid 1 = 0 \mid 1 \Rightarrow$

$$\begin{array}{r} 0\ 0 \\ \mid\ 0\ 1 \\ \hline 0\ 1 \end{array} \Rightarrow 1$$

$s = 3 \text{ OR } 1 \Rightarrow$

$$\begin{array}{r} 1\ 1 \\ \text{OR}\ 0\ 1 \\ \hline 1\ 1 \end{array} \Rightarrow \boxed{3}$$

A. 1 B. 2
C. 3 D. 5